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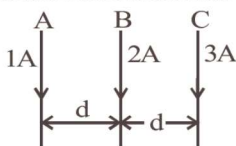
Moving Charges and Magnetism

Multiple Choice Questions (MCQs)

DIRECTIONS : This section contains multiple choice questions. Each question has four choices (a), (b), (c) and (d) out of which only one is correct.

- Magnetic field at the centre of a circular coil of radius r , through which a current I flows is
 - directly proportional to r
 - inversely proportional to I
 - directly proportional to I
 - directly proportional to I^2
- A long solenoid carrying a current produces a magnetic field B along its axis. If the current is double and the number of turns per cm is halved, the new value of the magnetic field is
 - $4B$
 - $B/2$
 - B
 - $2B$
- The magnetic field around a long straight current carrying wire is
 - spherical symmetry
 - cylindrical symmetry
 - cubical symmetry
 - unsymmetrical
- Magnetic field at the centre of a circular coil of radius r , through which a current I flows is
 - directly proportional to r
 - inversely proportional to I
 - directly proportional to I
 - directly proportional to I^2
- Which of the following statements is/are correct?
 - The magnetic field in the open space inside the toroid is constant.
 - The magnetic field in the open space exterior to the toroid is constant.
 - The magnetic field inside the core of toroid is constant.
 - I and II
 - II and III
 - III only
 - I only
- A particle of mass m and charge q enters a magnetic field B perpendicularly with a velocity v . The radius of the circular path described by it will be
 - Bq/mv
 - mq/Bv
 - mB/qv
 - mv/Bq
- Which of the following statement(s) is/are true/false?
 - A charged particle moves perpendicular to the magnetic field. Its kinetic energy remains constant, but momentum changes.
 - Force acts perpendicular to the velocity of particle.
 - F, F
 - F, T
 - T, T
 - T, F
- In a region, steady and uniform electric and magnetic fields are present. These two fields are parallel to each other. A charged particle is released from rest in this region. The path of the particle will be a
 - helix
 - straight line
 - ellipse
 - circle
- Direction of force due to magnetic field on a moving charged particle is
 - perpendicular to direction of velocity of charged particle.
 - perpendicular to direction of magnetic field.
 - parallel to direction of velocity of charged particle.
 - parallel to the direction of magnetic field.
 True/false statements are
 - T, F, F, T
 - T, T, F, F
 - T, F, T, F
 - F, F, T, T
- Consider the following statements and select the true/false.
 - Force on the charged particle will be zero if it is at rest.
 - Direction of force on moving charge particle is given by Fleming's Left Hand Rule.
 - A charged particle enters a region of uniform magnetic field at an angle of 45° to the magnetic lines of force, the path of the particle is a circle.
 - There is no change in the kinetic energy of a charged particle moving in a magnetic field although a magnetic force is acting on it.
 - F, F, F, T
 - F, T, F, F
 - T, T, F, T
 - F, F, T, T
- A beam of electrons is moving with constant velocity in a region having simultaneous perpendicular electric and magnetic fields of strength 20 Vm^{-1} and 0.5 T respectively at right angles to the direction of motion of the electrons. Then the velocity of electrons must be
 - 8 m/s
 - 20 m/s
 - 40 m/s
 - $\frac{1}{40} \text{ m/s}$
- Two thin, long, parallel wires, separated by a distance 'd' carry a current of 'i' A in the same direction. They will
 - repel each other with a force of $\mu_0 i^2 / (2\pi d)$
 - attract each other with a force of $\mu_0 i^2 / (2\pi d)$
 - repel each other with a force of $\mu_0 i^2 / (2\pi d^2)$
 - attract each other with a force of $\mu_0 i^2 / (2\pi d^2)$

13. Three wires A, B and C are situated at the same distance. A current of 1A, 2A, 3A flows through these wires in the same direction. Then the resultant force on B is directed
- Towards A
 - Towards C
 - Perpendicular to the plane of paper and outward
 - Perpendicular to the plane of paper and inward



14. The magnetic dipole moment of a current loop is independent of
- magnetic field in which it is lying
 - number of turns
 - area of the loop
 - current in the loop
15. A current carrying loop is placed in a uniform magnetic field. The torque acting on it does not depend upon
- shape of the loop
 - area of the loop
 - value of the current
 - magnetic field
16. A square current carrying loop is suspended in a uniform magnetic field acting in the plane of the loop. If the force on one arm of the loop is $\vec{F}$, the net force on the remaining three arms of the loop is
- $3\vec{F}$
 - $-\vec{F}$
 - $-3\vec{F}$
 - $\vec{F}$
17. The current sensitivity of a galvanometer is defined as
- the current flowing through the galvanometer when a unit voltage is applied across its terminals.
 - current per unit deflection.
 - deflection per unit current.
 - deflection per unit current when a unit voltage is applied across its terminals.
18. The deflection in a moving coil galvanometer is
- directly proportional to the torsional constant
 - directly proportional to the number of turns in the coil
 - inversely proportional to the area of the coil
 - inversely proportional to the current flowing in g
19. The galvanometer cannot be used as an ammeter to measure the value of current in a given circuit. The following reasons are
- galvanometer gives full scale deflection for a small current.
 - galvanometer has a large resistance.
 - a galvanometer can give inaccurate values.
- The correct reasons are:
- I and II
 - II and III
 - I and III
 - I, II and III
20. A galvanometer having a resistance of 80 ohms is shunted by a wire of resistance 2 ohms. If the total current is 1 amp., the part of it passing through the shunt will be
- 0.25 amp
 - 0.8 amp
 - 0.02 amp
 - 0.5 amp
21. The resistance of an ammeter is $13\ \Omega$ and its scale is graduated for a current upto 100 amps. After an additional shunt has been connected to this ammeter it becomes possible to measure currents upto 750 amperes by this meter. The value of shunt-resistance is
- $2\ \Omega$
 - $0.2\ \Omega$
 - $2\text{ k}\Omega$
 - $20\ \Omega$

22. The magnetic dipole moment of a current carrying coil does **not** depend upon [CBSE 2020]

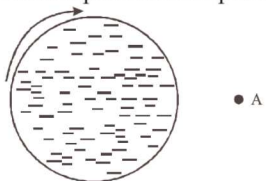
- number of turns of the coil.
 - cross-sectional area of the coil.
 - current flowing in the coil.
 - material of the turns of the coil.
23. Two charged particles traverse identical helical paths in a completely opposite sense in a uniform magnetic field $\mathbf{B} = B_0 \hat{\mathbf{k}}$.
- They have equal z-components of momenta
 - They must have equal charges
 - They necessarily represent a particle, anti-particle pair
 - The charge to mass ratio satisfy

$$\left(\frac{e}{m}\right)_1 + \left(\frac{e}{m}\right)_2 = 0$$

24. Biot-Savart law indicates that the moving electrons (velocity $\mathbf{v}$) produce a magnetic field $\mathbf{B}$ such that
- $\mathbf{B}$ is perpendicular of $\mathbf{v}$
 - $\mathbf{B}$ is parallel to $\mathbf{v}$
 - it obeys inverse cube law
 - it is along the line joining the electron and point of observation.
25. A current carrying circular loop of radius R is placed in the x-y plane with centre at the origin. Half of the loop with $x > 0$ is now bent so that it now lies in the y-z plane.
- The magnitude of magnetic moment now diminishes
 - The magnetic moment does not change
 - The magnitude of $\mathbf{B}$ at $(0, 0, z)$, $z > R$ increases
 - The magnitude of $\mathbf{B}$ at $(0, 0, z)$, $z \gg R$ is unchanged
26. An electron is projected with uniform velocity along the axis of a current carrying long solenoid. Which of the following is true?
- The electron will be accelerated along the axis
 - The electron path will be circular about the axis
 - The electron will experience a force at 45° to the axis and hence execute a helical path
 - The electron will continue to move with uniform velocity along the axis of the solenoid
27. A galvanometer of resistance $100\ \Omega$ gives a full scale deflection for a current of 10^{-5} A . To convert it into an ammeter capable of measuring upto 1 A, we should connect a resistance of
- $1\ \Omega$ in parallel
 - $10^{-3}\ \Omega$ in parallel
 - $10^5\ \Omega$ in series
 - $100\ \Omega$ in series
28. An electric charge in uniform motion produces
- an electric field only
 - a magnetic field only
 - both electric and magnetic fields
 - no such field at all
29. An electron having a charge e moves with a velocity $\mathbf{v}$ in X-direction. Magnetic field acts on it in Y-direction? The force on the electron acts in
- positive direction of Y-axis
 - negative direction of Y-axis
 - positive direction of Z-axis
 - negative direction of Z-axis

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30. An electron moves in a circular orbit with a uniform speed v . It produces a magnetic field B at the centre of the circle. The radius of the circle is proportional to
- (a) $\sqrt{\frac{B}{v}}$ (b) $\frac{B}{v}$ (c) $\sqrt{\frac{v}{B}}$ (d) $\frac{v}{B}$
31. A charged particle of mass m and charge q travels on a circular path of radius r that is perpendicular to a magnetic field B . The time taken by the particle to complete one revolution is
- (a) $\frac{2\pi q^2 B}{m}$ (b) $\frac{2\pi m q}{B}$ (c) $\frac{2\pi m}{qB}$ (d) $\frac{2\pi q B}{m}$
32. A charge q is moving with a velocity v parallel to a magnetic field B . Force on the charge due to magnetic field is
- (a) $q v B$ (b) $q B/v$ (c) zero (d) $B v/q$
33. Consider a moving charged particle in a region of magnetic field. Which of the following statements are correct?
- (a) If v is parallel to B , then path of particle is spiral.
 (b) If v is perpendicular to B , then path of particle is a circle.
 (c) If v has a component along B , then path of particle is zig-zag.
 (d) If v is along B , then path of particle is a circle.
34. Which of the above statements is/are incorrect?
 A velocity selector; (a region of perpendicular electric and magnetic field)
- (a) allows charged particles to pass straight when $v = E/B$.
 (b) deflects particles in the direction of electric field when $v < E/B$.
 (c) deflects particles in a direction perpendicular to both v and B , when $v < E/B$.
 (d) deflects all particles in a direction perpendicular to both E and B .
35. Select the incorrect statement about Lorentz Force.
- (a) In presence of electric field $\vec{E}(r)$ and magnetic field $\vec{B}(r)$ the force on a moving electric charge is $\vec{F} = q[\vec{E}(r) + v \times \vec{B}(r)]$
 (b) The force, due to magnetic field on a negative charge is opposite to that on a positive charge.
 (c) The force due to magnetic field become zero if velocity and magnetic field are parallel or antiparallel.
 (d) For a static charge the magnetic force is maximum.
36. In the resonance condition
- (a) the frequency of revolution of charged particle is equal to the frequency of A.C. voltage sources
 (b) the frequency of revolution of charged particle is equal to the frequency of applied magnetic field
 (c) the frequency of revolution of charged particle is equal to the frequency of rotation of earth
 (d) the frequency of revolution of charged particle, frequency of A.C. source and frequency of magnetic field are equal
37. A charged particle enters in a uniform magnetic field with a certain velocity. The power delivered to the particle by the magnetic field depends on
- (a) force exerted by magnetic field and velocity of the particle.
 (b) angular speed ω and radius r of the circular path.
 (c) angular speed ω and acceleration of the particle.
 (d) None of these
38. A proton moving with a constant velocity passes through a region of space without any change in its velocity. If E and B represent the electric and magnetic fields respectively, this region of space may not have
- (a) $E = 0, B = 0$ (b) $E = 0, B \neq 0$
 (c) $E \neq 0, B = 0$ (d) $E \neq 0, B \neq 0$
39. A charged particle with charge q enters a region of constant, uniform and mutually orthogonal fields $\vec{E}$ and $\vec{B}$ with a velocity $\vec{v}$ perpendicular to both $\vec{E}$ and $\vec{B}$, and comes out without any change in magnitude or direction of $\vec{v}$. Then
- (a) $\vec{v} = \vec{B} \times \vec{E} / E^2$ (b) $\vec{v} = \vec{E} \times \vec{B} / B^2$
 (c) $\vec{v} = \vec{B} \times \vec{E} / B^2$ (d) $\vec{v} = \vec{E} \times \vec{B} / E^2$
40. If an electron and a proton having same momenta enter perpendicular to a magnetic field, then
- (a) curved path of electron and proton will be same (ignoring the sense of revolution)
 (b) they will move undeflected
 (c) curved path of electron is more curved than that of the proton
 (d) path of proton is more curved
41. A beam of electrons is moving with constant velocity in a region having simultaneous perpendicular electric and magnetic fields of strength 20 Vm^{-1} and 0.5 T respectively at right angles to the direction of motion of the electrons. Then the velocity of electrons must be
- (a) 8 m/s (b) 20 m/s (c) 40 m/s (d) $\frac{1}{40} \text{ m/s}$
42. A charge moving with velocity v in X -direction is subjected to a field of magnetic induction in negative X -direction. As a result, the charge will
- (a) remain unaffected
 (b) start moving in a circular path Y - Z plane
 (c) retard along X -axis
 (d) move along a helical path around X -axis
43. A proton (mass m and charge $+e$) and an α -particle (mass $4m$ and charge $+2e$) are projected with the same kinetic energy at right angles to the uniform magnetic field. Which one of the following statements will be true?
- (a) The α -particle will be bent in a circular path with a small radius that for the proton
 (b) The radius of the path of the α -particle will be greater than that of the proton
 (c) The α -particle and the proton will be bent in a circular path with the same radius
 (d) The α -particle and the proton will go through the field in a straight line

44. A positively charged particle moving into east enters a region of uniform magnetic field directed vertically upwards. The particle will
 (a) continue to move due east
 (b) move in a circular orbit with its speed unchanged
 (c) move in a circular orbit with its speed increased
 (d) gets deflected vertically upwards.
45. The negatively and uniformly charged nonconducting disc as shown is rotated clockwise. The direction of the magnetic field at point A in the plane of the disc is
 (a) into the page
 (b) out of the page
 (c) up the page
 (d) down the page
- 
46. A proton (mass = 1.67×10^{-27} kg and charge = 1.6×10^{-19} C) enters perpendicular to a magnetic field of intensity 2 weber/m² with a velocity 3.4×10^7 m/sec. The acceleration of the proton should be
 (a) 6.5×10^{15} m/sec²
 (b) 6.5×10^{13} m/sec²
 (c) 6.5×10^{11} m/sec²
 (d) 6.5×10^9 m/sec²
47. The magnetic field around a long straight current carrying wire is
 (a) spherical symmetry
 (b) cylindrical symmetry
 (c) cubical symmetry
 (d) unsymmetrical
48. A current is passed through a straight wire. The magnetic field established around it has its lines of force
 (a) circular and endless
 (b) oval in shape and endless
 (c) straight
 (d) All of the above
49. A current carrying coil is subjected to a uniform magnetic field. The coil will orient so that its plane becomes
 (a) inclined at 45° to the magnetic field
 (b) inclined at any arbitrary angle to the magnetic field
 (c) parallel to the magnetic field
 (d) perpendicular to the magnetic field
50. A small current element of length dl and carrying current is placed at (1, 1, 0) and is carrying current in '+z' direction. If magnetic field at origin be $\vec{B}_1$ and at point (2, 2, 0) be $\vec{B}_2$ then
 (a) $\vec{B}_1 = \vec{B}_2$
 (b) $|\vec{B}_1| = |2\vec{B}_2|$
 (c) $\vec{B}_1 = -\vec{B}_2$
 (d) $\vec{B}_1 = -2\vec{B}_2$
51. A current I flows along the length of an infinitely long, straight, thin walled pipe. Then
 (a) the magnetic field at all points inside the pipe is the same, but not zero
 (b) the magnetic field is zero only on the axis of the pipe
 (c) the magnetic field is different at different points inside the pipe
 (d) the magnetic field at any point inside the pipe is zero
52. If a copper rod carries a direct current, the magnetic field associated with the current will be
 (a) only inside the rod
 (b) only outside the rod
 (c) both inside and outside the rod
 (d) neither inside nor outside the rod
53. Biot-Savart law indicates that the moving electrons velocity (V) produce a magnetic field B such that
 (a) $B \parallel V$
 (b) $B \perp V$
 (c) it obeys inverse cube law
 (d) it is along the line joining electron and point of observation
54. Ampere's circuital law is equivalent to
 (a) Biot-Savart law
 (b) Coulomb's law
 (c) Faraday's law
 (d) Kirchhoff's law
55. A solenoid of length 1.5 m and 4 cm diameter possesses 10 turns per cm. A current of 5 A is flowing through it, the magnetic induction at axis inside the solenoid is ($\mu_0 = 4\pi \times 10^{-7}$ weber amp⁻¹ m⁻¹)
 (a) $4\pi \times 10^{-5}$ gauss
 (b) $2\pi \times 10^{-5}$ gauss
 (c) $4\pi \times 10^{-5}$ tesla
 (d) $2\pi \times 10^{-5}$ tesla
56. The magnetic field B at a point on one end of a solenoid having n turns per metre length and carrying a current of i ampere is given by
 (a) $\frac{\mu_0 n i}{e}$
 (b) $\frac{1}{2} \mu_0 n i$
 (c) $4\pi \mu_0 n i$
 (d) $n i$
57. A long solenoid has 200 turns per cm and carries a current i . The magnetic field at its centre is 6.28×10^{-2} Weber/m². Another long solenoid has 100 turns per cm and it carries a current $\frac{i}{3}$. The value of the magnetic field at its centre is
 (a) 1.05×10^{-2} Weber/m²
 (b) 1.05×10^{-5} Weber/m²
 (c) 1.05×10^{-3} Weber/m²
 (d) 1.05×10^{-4} Weber/m²
58. Two concentric circular coils of ten turns each are situated in the same plane. Their radii are 20 and 40 cm and they carry respectively 0.2 and 0.4 ampere current in opposite direction. The magnetic field in weber/m² at the centre is
 (a) $\mu_0/80$
 (b) $7\mu_0/80$
 (c) $(5/4)\mu_0$
 (d) zero
59. A coil of one turn is made of a wire of certain length and then from the same length a coil of two turns is made. If the same current is passed in both the cases, then the ratio of the magnetic inductions at their centres will be
 (a) 2 : 1
 (b) 1 : 4
 (c) 4 : 1
 (d) 1 : 2
60. Magnetic field intensity at the centre of a coil of 50 turns, radius 0.5 m and carrying a current of 2 A is
 (a) 0.5×10^{-5} T
 (b) 1.25×10^{-4} T
 (c) 3×10^{-5} T
 (d) 4×10^{-5} T
61. A coil carrying electric current is placed in uniform magnetic field, then
 (a) torque is formed
 (b) e.m.f is induced
 (c) both (a) and (b) are
 (d) None of these
62. Two long wires are hanging freely. They are joined first in parallel and then in series and then are connected with a battery. In both cases which type of force acts between the two wires?
 (a) Attraction force when in parallel and repulsion force when in series
 (b) Repulsion force when in parallel and attraction force when in series
 (c) Repulsion force in both cases
 (d) Attraction force in both cases

Moving Charges and Magnetism

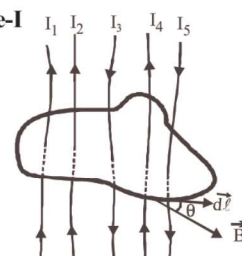
63. A moving coil galvanometer has N number of turns in a coil of effective area A , it carries a current I . The magnetic field B is radial. The torque acting on the coil is
 (a) NA^2B^2I (b) $NABI^2$ (c) N^2ABI (d) $NABI$
64. A conducting circular loop of radius r carries a constant current i . It is placed in a uniform magnetic field $\vec{B}_0$ such that $\vec{B}_0$ is perpendicular to the plane of the loop. The magnetic force acting on the loop is
 (a) irB_0 (b) $2\pi irB_0$ (c) zero (d) πirB_0
65. Two long conductors, separated by a distance d carry current I_1 and I_2 in the same direction. They exert a force F on each other. Now the current in one of them is increased to two times and its direction is reversed. The distance is also increased to $3d$. The new value of the force between them is
 (a) $-\frac{2F}{3}$ (b) $\frac{F}{3}$ (c) $-2F$ (d) $-\frac{F}{3}$
66. A current of 5 ampere is flowing in a wire of length 1.5 metres. A force of 7.5 N acts on it when it is placed in a uniform magnetic field of 2 tesla. The angle between the magnetic field and the direction of the current is
 (a) 30° (b) 45° (c) 60° (d) 90°
67. A straight wire of length 0.5 metre and carrying a current of 1.2 ampere is placed in uniform magnetic field of induction 2 tesla. The magnetic field is perpendicular to the length of the wire. The force on the wire is
 (a) 2.4 N (b) 1.2 N (c) 3.0 N (d) 2.0 N
68. An electric current of 30 ampere is flowing in each of two parallel conducting wires placed 5 cm apart. The force acting per unit length on either of the wires will be
 (a) 3.6×10^{-3} N/m (b) 3.6×10^{-3} dyne/cm
 (c) 3.6×10^{-5} N/m (d) 3.6×10^{-2} N/m
69. The distance between the wires of electric mains is 12 cm. These wires experience 4 mg wt. per unit length. The value of current flowing in each wire will be
 (a) 4.85 A (b) 0
 (c) 4.85×10^{-2} A (d) 4.85×10^{-4} A
70. A current carrying conductor placed in a magnetic field experiences maximum force when angle between current and magnetic field is
 (a) $3\pi/4$ (b) $\pi/2$ (c) $\pi/4$ (d) zero
71. A current of 3 A is flowing in a linear conductor having a length of 40 cm. The conductor is placed in a magnetic field of strength 500 gauss and makes an angle of 30° with the direction of the field. It experiences a force of magnitude
 (a) 3×10^{-4} N (b) 3×10^{-2} N
 (c) 3×10^2 N (d) 3×10^4 N
72. Two straight horizontal parallel wires are carrying the same current in the same direction, d is the distance between the wires. You are provided with a small freely suspended magnetic needle. At which of the following positions will the orientation of the needle be independent of the magnitude of the current in the wires?
 (a) At a distance $d/2$ from any of the wires
 (b) At a distance $d/2$ from any of the wires in the horizontal plane
 (c) Anywhere on the circumference of a vertical circle of radius d and centre halfway between the wires
 (d) At points halfway between the wires in the horizontal plane
73. Two long parallel wires are at a distance of 1 metre. Both of them carry 5 ampere of current. The force of attraction per unit length between the two wires is
 (a) 50×10^{-7} N/m (b) 2×10^{-8} N/m
 (c) 5×10^{-8} N/m (d) 10^{-7} N/m
74. A and B are two conductors carrying a current i in the same direction. x and y are two electron beams moving in the same direction. Then
 (a) there will be repulsion between A and B , attraction between x and y
 (b) there will be attraction between A and B , repulsion between x and y
 (c) there will be repulsion between A and B and also x and y
 (d) there will be attraction between A and B and also x and y

Case/Passage Based Questions

DIRECTIONS : Study the given paragraph(s) and answer the following questions.

Passage/Case-I

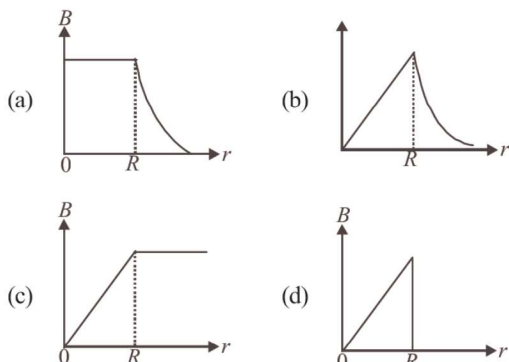
Ampere's circuital law states that the line integral of magnetic field induction $\vec{B}$ around any closed path in vacuum is equal to μ_0 times the total current crossing the area bounded by the closed path provided the electric field inside the loop remains constant.



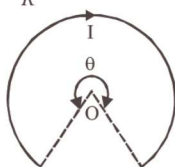
i.e., $\oint \vec{B} \cdot d\vec{\ell} = \mu_0 \sum I$ here $\sum I = I_1 + I_2 - I_3 + I_4 - I_5$

75. The magnetic field B at a point on one end of a solenoid having n turns per metre length and carrying a current of i ampere is given by
 (a) $\frac{\mu_0 ni}{e}$ (b) $\frac{1}{2} \mu_0 ni$
 (c) $4\pi \mu_0 ni$ (d) ni
76. If a long hollow copper pipe carries a direct current, the magnetic field associated with the current will be
 (a) only inside the pipe
 (b) only outside the pipe
 (c) neither inside nor outside the pipe
 (d) both inside and outside the pipe

77. The correct plot of the magnitude of magnetic field $\vec{B}$ vs distance r from centre of the wire is, if the radius of wire is R



78. A current of I ampere flows in a wire forming a circular arc of radius r metres subtending an angle θ at the centre as shown. The magnetic field at the centre O in tesla is



- (a) $\frac{\mu_0 I \theta}{4\pi r}$ (b) $\frac{\mu_0 I \theta}{2\pi r}$ (c) $\frac{\mu_0 I \theta}{2r}$ (d) $\frac{\mu_0 I \theta}{4r}$

79. The figure shows n (n being an even number) wires placed along the surface of a cylinder of radius r . Each wire carries current i in the same direction. The net magnetic field on the axis of the cylinder is



- (a) $\mu_0 ni$ (b) $\frac{\mu_0 ni}{2\pi r}$ (c) zero (d) $\frac{\mu_0 ni}{4\pi r}$

Passage/Case-II

The moving charged particle will experience **electric force** $\vec{F}_e = q\vec{E}$ and **magnetic force** $\vec{F}_m = q(\vec{v} \times \vec{B})$

Net force on the charged particle $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$ "Lorentz-force"

Depending on the direction of $\vec{v}$, $\vec{E}$ and $\vec{B}$ various situations are possible and the motion in general is quite complex.

80. An electron having a charge e moves with a velocity v in X-direction. An electric field acts on it in Y-direction? The force on the electron acts in
- positive direction of Y-axis
 - negative direction of Y-axis
 - positive direction of Z-axis
 - negative direction of Z-axis
81. Lorentz force is
- electrostatic force acting on a charged particle.
 - magnetic force acting on a moving charged particle.
 - the vector sum of electrostatic and magnetic force acting on a moving charged particle.
 - the vector sum of gravitational and magnetic force acting on a moving charged particle.

82. A certain region has an electric field $\vec{E} = (2\hat{i} - 3\hat{j})$ N/C and a uniform magnetic field $\vec{B} = (5\hat{i} + 3\hat{j} + 4\hat{k})$ T. The force experienced by a charge $1C$ moving with velocity $(\hat{i} + 2\hat{j})$ ms^{-1} is

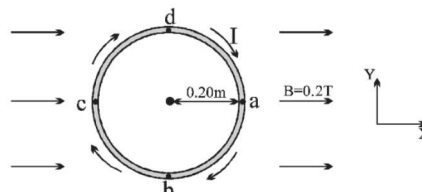
- (a) $(10\hat{i} - 7\hat{j} - 7\hat{k})$ (b) $(10\hat{i} + 7\hat{j} + 7\hat{k})$
(c) $(-10\hat{i} + 7\hat{j} + 7\hat{k})$ (d) $(10\hat{i} + 7\hat{j} - 7\hat{k})$

83. A proton moving with a constant velocity passes through a region of space without any change in its velocity. If $\vec{E}$ and $\vec{B}$ represent the electric and magnetic fields respectively, this region of space may not have
- $\vec{E} = 0, \vec{B} = 0$
 - $\vec{E} = 0, \vec{B} \neq 0$
 - $\vec{E} \neq 0, \vec{B} = 0$
 - $\vec{E} \neq 0, \vec{B} \neq 0$

84. A charged particle with velocity 2×10^3 m/s passes undeflected through electric and magnetic field. Magnetic field is 1.5 tesla. The electric field intensity would be
- 2×10^3 N/C
 - 1.5×10^3 N/C
 - 3×10^3 N/C
 - $4/3 \times 10^{-3}$ N/C

Passage/Case-III

A rigid circular loop has a radius of 0.20 m and is in the x-y plane. A clockwise current I is carried by the loop, as shown. The magnitude of the magnetic moment of the loop is 0.75 A-m^2 . A uniform external magnetic field, $B = 0.20 \text{ T}$ in the positive x-direction, is present



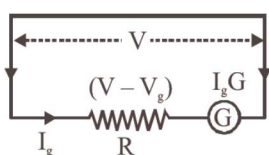
85. In figure, the magnitude of the magnetic torque exerted on the loop is closest to
- 0.55 N-m
 - 0.15 N-m
 - 0.45 N-m
 - 0.35 N-m
86. In figure the loop is released from rest. The initial motion of the loop is described by
- point a moves out of the plane, point c moves into the plane
 - points a, b, c and d move counterclockwise
 - point a, b, c and d move clockwise
 - point c moves out of the plane, point a moves into the plane
87. In figure, an external torque changes the orientation of loop from one of lowest potential energy to one of highest potential energy. The work done by the external torque is closest to
- 0.5 J
 - 0.2 J
 - 0.3 J
 - 0.4 J
88. A wire is placed parallel to the lines of force in a magnetic field and a current flows in the wire. Then
- the wire will experience a force in the direction of the magnetic field
 - the wire will not experience any force at all
 - the wire will experience a force in a direction opposite to the field
 - it experiences a force in a direction perpendicular to lines of force

Moving Charges and Magnetism

89. A current carrying coil is subjected to a uniform magnetic field. The coil will orient so that its plane becomes
- inclined at 45° to the magnetic field
 - inclined at any arbitrary angle to the magnetic field
 - parallel to the magnetic field
 - perpendicular to the magnetic field

Passage/Case-IV

- (a) A galvanometer is converted to voltmeter by putting a high resistance in series with it.
Total resistance of voltmeter = $G + R$ where G is the galvanometer resistance.
 R is the resistance added in series.



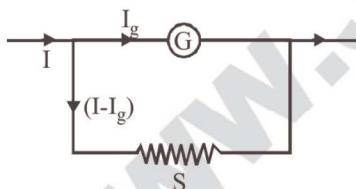
Current through the galvanometer, $I_g = \frac{V}{G + R}$

where V is the potential difference across the voltmeter.

And resistance, $R = \frac{V}{I_g} - G$

- (b) A galvanometer is converted into an ammeter by connecting a low resistance (shunt) in parallel with it.

Shunt $S = \left(\frac{I_g}{I - I_g} \right) G$ where G is the galvanometer resistance.



I is the total current through the ammeter.

I_g is the current through the ammeter.

Effective resistance of the ammeter $R = \frac{GS}{G + S}$

The range of an ammeter can be increased N times by reducing shunt $S = G/N - 1$

An ideal ammeter has zero resistance.

90. A galvanometer of resistance, G is shunted by a resistance S ohm. To keep the main current in the circuit unchanged, the resistance to be put in series with the galvanometer is

(a) $\frac{S^2}{(S + G)}$ (b) $\frac{SG}{(S + G)}$ (c) $\frac{G^2}{(S + G)}$ (d) $\frac{G}{(S + G)}$

91. A galvanometer having a coil resistance of 60Ω shows full scale deflection when a current of 1.0 amp passes through it. It can be converted into an ammeter to read currents upto 5.0 amp by

- (a) putting in series a resistance of 15Ω

- (b) putting in series a resistance of 240Ω
(c) putting in parallel a resistance of 15Ω
(d) putting in parallel a resistance of 240Ω

92. A milli voltmeter of 25 milli volt range is to be converted into an ammeter of 25 ampere range. The value (in ohm) of necessary shunt will be

- (a) 0.001 (b) 0.01 (c) 1 (d) 0.05

93. A moving coil galvanometer of resistance 100Ω is used as an ammeter using a resistance 0.1Ω . The maximum deflection current in the galvanometer is $100 \mu A$. Find the minimum current in the circuit so that the ammeter shows maximum deflection

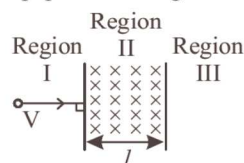
- (a) 100.1 mA (b) 1000.1 mA
(c) 10.01 mA (d) 1.01 mA

94. A galvanometer of resistance 5 ohms gives a full scale deflection for a potential difference of 10 mV. To convert the galvanometer into a voltmeter giving a full scale deflection for a potential difference of 1 V, the size of the resistance that must be attached to the voltmeter is

- (a) 0.495 ohm (b) 49.5 ohm
(c) 495 ohm (d) 4950 ohm

Passage/Case-V

A particle of mass m and charge q , moving with velocity V enters Region II normal to the boundary as shown in the figure. Region II has a uniform magnetic field B perpendicular to the plane of the paper. The length of the Region II is l .



95. (a) The particle enters Region III only if its velocity

$V > \frac{qB}{m}$

- (b) The particle enters Region III only if its velocity

$V < \frac{qB}{m}$

- (c) The particle enters Region III only if its velocity

$V = \frac{qB}{m}$

- (d) All of the above

96. Path length of the particle in Region II is maximum when

(a) velocity $V = \frac{qB}{2m}$ (b) velocity $V = \frac{2qB}{m}$

(c) velocity $V = \frac{qB}{m}$ (d) velocity $V = \frac{4qB}{m}$

97. Time spent in Region II as long as the particle returns to Region I is

- (a) two times if velocity V is doubled

- (b) halved if velocity is doubled

- (c) halved if velocity is halved

- (d) same for any value of V

98. If the direction of the initial velocity of the charged particle is neither along nor perpendicular to that of the magnetic field, then the orbit will be
 (a) a straight line (b) an ellipse
 (c) a circle (d) a helix
99. A charged particle moves with velocity $\vec{v}$ in a uniform magnetic field $\vec{B}$. The magnetic force experienced by the particle is
 (a) always zero
 (b) never zero
 (c) zero, if $\vec{B}$ and $\vec{v}$ are perpendicular
 (d) zero, if $\vec{B}$ and $\vec{v}$ are parallel

» Assertion & Reason

DIRECTIONS : Each of these questions contains an assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are **correct** and the Reason is the **correct explanation** of the Assertion.
 (b) If both **Assertion** and **Reason** are correct but Reason is **not the correct explanation** of the Assertion.
 (c) If the **Assertion** is **correct** but **Reason** is **incorrect**.
 (d) If the **Assertion** is **incorrect** but the **Reason** is **correct**.
100. **Assertion:** Ampere's circuital law is analogous of Biot-Savart's law.
Reason: Ampere's circuital law cannot be derived from the Biot-savart's law.
101. **Assertion :** Figure shows a current carrying circular loop. The magnetic field at the centre of loop is zero.

Reason : Magnitude of magnetic field at the centre of circular loop carrying current i is given by $B = \frac{\mu_0 n i}{R}$.
102. **Assertion :** A current I flows along the length of an infinitely long straight and thin walled pipe. Then the magnetic field at any point inside the pipe is zero.
Reason : $\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I$ and $\sum I_{in} = 0$
103. **Assertion :** The magnetic field at the end of a very long current carrying solenoid is half of that at the center.
Reason : If the solenoid is sufficiently long, the field within it is uniform.
104. **Assertion :** Magnetic field interacts with a moving charge and not with a stationary charge.
Reason : A moving charge produces a magnetic field.
105. **Assertion :** If a charged particle is released from rest in a region of uniform electric and magnetic fields parallel to each other, it will move in a straight line.
Reason : The electric field exerts no force on the particle but the magnetic field does.

106. **Assertion :** A charged particle moves in a uniform magnetic field. The velocity of the particle at some instant makes an acute angle with the magnetic field. The path of the particle is a helix with constant pitch.

Reason : The force on the particle is given by $\vec{F} = q(\vec{v} \times \vec{B})$.

107. **Assertion:** The work done by magnetic force on a moving charged particle is zero.
Reason: The work done by magnetic force on a charged particle is zero as the force is always parallel to velocity of particle.
108. **Assertion :** A direct current flows through a thin conductor produces magnetic field only outside the conductor.
Reason : There is no flow of charge carriers inside the conductor.
109. **Assertion :** The net charge in a current carrying wire is zero and so magnetic force on the wire in magnetic field is zero.

Reason : The force on a current carrying wire is given by $F = qVB \sin\theta$ [where q = charge, V = potential difference, B = field, θ = angle]

110. **Assertion:** Electron moving perpendicular to $\vec{B}$ will perform circular motion ?

Reason: Force by magnetic field is perpendicular to velocity

111. **Assertion :** If the current in a solenoid is reversed in direction while keeping the same magnitude, the magnetic field energy stored in the solenoid remains unchanged.

Reason : Magnetic field energy density is proportional to the magnetic field.

» Match the Following

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (1, 2, 3, 4) in column-II.

112. Match Column I and Column II.

Column I	Column II
(A) Biot-Savart's law	(1) $\frac{\mu_0 I_1 I_2}{2\pi d}$
(B) Ampere's circuital law	(2) $q[\vec{E} + (\vec{v} \times \vec{B})]$
(C) Force between two parallel current carrying conductors	(3) $\oint \vec{B} \cdot d\vec{\ell} = \mu_0 \sum i$
(D) Lorentz force	(4) $\vec{B} = \frac{\mu_0 i}{4\pi} \int \frac{d\vec{\ell} \sin\theta}{r^2} \hat{n}$
(a) (A) $\rightarrow$ (4); (B) $\rightarrow$ (3); (C) $\rightarrow$ (1); (D) $\rightarrow$ (2)	
(b) (A) $\rightarrow$ (2); (B) $\rightarrow$ (2); (C) $\rightarrow$ (4); (D) $\rightarrow$ (3)	
(c) (A) $\rightarrow$ (4); (B) $\rightarrow$ (3); (C) $\rightarrow$ (2); (D) $\rightarrow$ (1)	
(d) (A) $\rightarrow$ (2); (B) $\rightarrow$ (1); (C) $\rightarrow$ (4); (D) $\rightarrow$ (3)	

» Fill in the Blanks

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

113. Lorentz force is _____ charged particle.
114. An electric charge $+q$ moves with velocity $\vec{v} = 3\hat{i} + 4\hat{j} + \hat{k}$ in an electromagnetic field given by $\vec{E} = 3\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{B} = \hat{i} + \hat{j} - 3\hat{k}$. The y -component of the force experienced by $+q$ is _____.
115. If m is magnetic moment and B is the magnetic field, then the torque is given by _____.
116. An 8 cm long wire carrying a current of 10 A is placed inside a solenoid perpendicular to its axis. If the magnetic field inside the solenoid is 0.3 T, then magnetic force on the wire is _____.
117. The magnetic induction at a point P which is at a distance of 4 cm from a long current carrying wire is 10^{-3} T. The field of induction at a distance 12 cm from the current will be _____ tesla.

» True / False

DIRECTIONS : Read the following statements and write your answer as true or false.

118. A charged particle undergoes uniform circular motion in a uniform magnetic field. The only force acting on the particle is that exerted by the uniform magnetic field. If now the speed of the same particle is somehow doubled keeping its charge and external magnetic field constant, then the centripetal force on the particle becomes four times.
119. The magnitude of centripetal force on a particle of mass m moving in a circle of radius R with uniform speed v is mv^2/R .
120. The magnetic field on the axis of a solenoid is non-uniform.
121. When a charged particle is projected along the axis of current carrying solenoid, the path of the particle is not changed.
122. A current carrying closed loop remains in equilibrium in a uniform and constant magnetic field parallel to its axis. Consider forces only due to this magnetic field.
123. Torque on a current carrying closed loop due to a magnetic field is maximum when the plane of the coil is parallel to the direction of the magnetic field.

ANSWER KEY & SOLUTIONS

1. (c) Field at the center of a circular coil of radius r is

$$B = \frac{\mu_0 I}{2r}$$

2. (c) $B = \mu_0 N_0 i$; $B_1 = (\mu_0) \left(\frac{N_0}{2} \right) (2i) = \mu_0 N_0 i = B$
 $\Rightarrow B_1 = B$

3. (b) Magnetic field is given by $B = \frac{\mu_0 i}{2\pi r}$ i.e., $B \propto \frac{1}{r}$ which implies that field has cylindrical symmetry.

4. (c) Field at the center of a circular coil of radius r is

$$B = \frac{\mu_0 I}{2r}$$

5. (c)

6. (d) Force, $F = qVB = \frac{mv^2}{R}$ $\therefore R = \frac{mv}{Bq}$

7. (c) A force which always perpendicular to velocity of the particle does no work on the particle, but changes the direction of momentum of the particle.

8. (b) The charged particle will move along the lines of electric field (and magnetic field). Magnetic field will exert no force. The force by electric field will be along the lines of uniform electric field. Hence the particle will move in a straight line.

9. (b) (i) Lorentz force depends on q , v and B (charge of the particle, the velocity and the magnetic field). Force on a negative charge is opposite to that on a positive charge.
 (ii) The magnetic force $q[\mathbf{v} \times \mathbf{B}]$ includes a vector product of velocity and magnetic field. The vector product makes the force due to magnetic field vanish (become zero) if velocity and magnetic field are parallel or anti-parallel. The force acts in a (sideways) direction perpendicular to both the velocity and the magnetic field. Its direction is given by the screw rule.

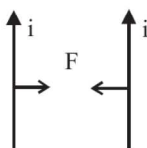
10. (c) When charged particle enters perpendicularly in a magnetic field, it moves in a circular path with a constant speed. Hence its kinetic energy also remains constant.

11. (c) The electron moves with constant velocity without deflection. Hence, force due to magnetic field is equal and opposite to force due to electric field.

$$qvB = qE \Rightarrow v = \frac{E}{B} = \frac{20}{0.5} = 40 \text{ m/s}$$

12. (b) When the direction of currents are same, the force is attractive. Hence, both the wires will attract each other Force per unit length.

$$\frac{F}{\ell} = \frac{\mu_0 i_1 i_2}{2\pi d} = \frac{\mu_0 i^2}{2\pi d}$$



13. (b) $F \propto i_1 i_2$, so force on B due to C will be greater than that due to A. Hence net force on B acts towards C.

14. (a) Current loop acts as a magnetic dipole. Its magnetic moment is given by

$$M = NIA$$

where N = number of turns, I = current in a loop, A = area of the loop

From the above relation, we can conclude that magnetic dipole moment of a current loop is independent of magnetic field in which it is lying.

15. (a) Because $\tau = NiAB \cos\theta$

16. (b) The force on the two arms parallel to the field is zero.



$\therefore$ Force on remaining arms $= -F$

17. (c) The current sensitivity of a galvanometer is defined as the deflection produced in the galvanometer per unit current flowing through it.

18. (b) $\theta = \frac{NiAB}{C} \Rightarrow \theta \propto N$ [Number of turns]

19. (a) The galvanometer cannot as such be used as an ammeter to measure the value of the current in a given circuit. This is for two reasons (i) Galvanometer is a very sensitive device, it gives a full-scale deflection for a current of the order of μA . (ii) For measuring currents, the galvanometer has to be connected in series and as it has a large resistance, this will change the value of the current in the circuit.

20. (c) [Hint $\Rightarrow S \times (I - I_g) = R_g \times I_g$]

21. (a) We know

$$\frac{I}{I_g} = 1 + \frac{G}{S} \quad \frac{750}{100} = 1 + \frac{13}{S} \quad S \Rightarrow 2\Omega$$

22. (d)

23. (d) As we know that the uniqueness of helical path is determined by its pitch

$$P(\text{Pitch}) = \frac{2\pi mv \cos\theta}{Bq}$$

Where θ is angle of velocity of charge particle with x-axis For the given pitch d correspond to charge particle, we have

$$\frac{q}{m} = \frac{2\pi v \cos\theta}{BP} = \text{constant}$$

Moving Charges and Magnetism

If motion is not helical, ($\theta = 0$)

As charged particles traverse identical helical paths in a completely opposite direction in a same magnetic field $\mathbf{B}$, LHS for two particles should be same and of opposite sign.

$$\therefore \left(\frac{e}{m}\right)_1 + \left(\frac{e}{m}\right)_2 = 0$$

24. (a) By Biot-Savart law

$$dB = \frac{Id \sin \theta}{r^2} = \left(\frac{I \times dl}{r} \right)$$

In Biot-Savart's law, magnetic field $\mathbf{B} \parallel dl \times \mathbf{r}$ and idl due to flow of electron is in opposite direction of $\mathbf{v}$ and by direction of cross product of two vectors

$$\mathbf{B} \perp \mathbf{V}$$

So, the magnetic field is $\perp$ to the direction of flow of charge.

25. (a) As the direction of magnetic moment of circular loop of radius R placed in the x - y plane is along z -direction and given by $M = I(\pi r^2)$, when half of the loop with $x > 0$ is now bent so that it now lies in the y - z plane, the magnitudes of magnetic field moment of each semicircular loop of radius R lie in the x - y plane and the y - z plane is $M' = I(\pi r^2)/4$ and the direction of magnetic field moments are along z -direction and x -direction respectively.

Then resultant is :

$$M_{\text{net}} = \sqrt{M'^2 + M'^2} = \sqrt{2} M' = \sqrt{2} I(\pi r^2)/4$$

So, $M_{\text{net}} < M$ or M diminishes.

Hence, the magnitude of magnetic moment is now diminishes.

26. (d) Magnetic Lorentz force :

$$F = qVB \sin \theta$$

Magnetic Lorentz force electron is projected with uniform velocity along the axis of a current carrying long solenoid $F = -qvB \sin 180^\circ = 0$ ($\theta = 0^\circ$) as magnetic field and velocity are parallel and electric field is zero ($E = 0$) due to this magnetic field (B) perpendicular to the direction of motion (V). So it will not affect the velocity of moving charge particle. So the electron will continue to move with uniform velocity along the axis of the solenoid

27. (b) Here, $R_g = 100 \Omega$; $I_g = 10^{-5} \text{ A}$; $I = 1 \text{ A}$; $S = ?$

$$S = \frac{I_g R_g}{I - I_g} = \frac{10^{-5} \times 100}{1 - 10^{-5}} = 10^{-3} \Omega \text{ in parallel}$$

28. (c) An electric charge in uniform motion produces both electric and magnetic fields.

29. (d)

30. (d) $r = \frac{mv}{qB} \Rightarrow r \propto \frac{v}{B}$

31. (c) Equating magnetic force to centripetal force,

$$\frac{mv^2}{r} = qvB \sin 90^\circ$$

$$\text{Time to complete one revolution, } T = \frac{2\pi r}{v} = \frac{2\pi m}{qB}$$

32. (c)

33. (b) If $\mathbf{v} \perp \mathbf{B}$, then path is circular and if $\mathbf{v}$ has a component along $\mathbf{B}$, then path will be helical.

34. (c) In a velocity selector, where F_e and F_B are electric and magnetic field, we get

$$\text{Case I } F_e = F_B \Rightarrow v = \frac{E}{B}$$

$$\text{Case II } F_e > F_B \Rightarrow v < \frac{E}{B}$$

$$\text{Case III } F_e < F_B \Rightarrow v > \frac{E}{B}$$

35. (d) If charge is not moving then the magnetic force is zero.

$$\text{Since } \vec{F}_m = q(\vec{v} \times \vec{B})$$

As $\vec{v} = 0$, for stationary charge

$$\therefore \vec{F}_m = 0$$

36. (a)

37. (d) Power = $\frac{\text{work done}}{\text{time}}$

As no work is done by magnetic force on the charged particle because magnetic force is perpendicular to velocity, hence power delivered is zero.

38. (c)

39. (b) Here, $\vec{E}$ and $\vec{B}$ are perpendicular to each other and the velocity $\vec{v}$ does not change; therefore

$$qE = qvB \Rightarrow v = \frac{E}{B}$$

If velocity $\vec{v}$ is $\perp$ to both $\vec{E}$ and $\vec{B}$,

$$\text{Also, } \left| \frac{\vec{E} \times \vec{B}}{B^2} \right| = \frac{E B \sin \theta}{B^2} = \frac{E B \sin 90^\circ}{B^2} = \frac{E}{B} = |\vec{v}| = v$$

40. (a) $r = mv/Bq$ is same for both.

41. (c) As electron move with constant velocity without deflection. Hence, force due to magnetic field is equal and opposite to force due to electric field.

$$qvB = qE \Rightarrow v = \frac{E}{B} = \frac{20}{0.5} = 40 \text{ m/s}$$

42. (a) The force acting on a charged particle in magnetic field is given by $\vec{F} = q(\vec{v} \times \vec{B})$ or $F = qvB \sin \theta$, When angle between $\mathbf{v}$ and $\mathbf{B}$ is 180° , $F = 0$

43. (c) $r = \frac{\sqrt{2mK}}{qB}$ i.e. $r \propto \frac{\sqrt{m}}{q}$

Here kinetic energy K and B are same.

$$\therefore \frac{r_p}{r_\alpha} = \frac{\sqrt{m_p}}{\sqrt{m_\alpha}} \cdot \frac{q_\alpha}{q_p} = \frac{\sqrt{m_p}}{\sqrt{4m_p}} \cdot \frac{2q_p}{q_p} = 1$$

44. (b) In a perpendicular magnetic field, the path of a charged particle is a circle, and the magnetic field does not cause any change in energy.

45. (a)

$$46. (a) \quad F = ma = qvB \Rightarrow a = \frac{qvB}{m}$$

$$= \frac{1.6 \times 10^{-19} \times 2 \times 3.4 \times 10^7}{1.67 \times 10^{-27}} = 6.5 \times 10^{15} \text{ m/sec}^2$$

47. (b) Magnetic field is given by $B = \frac{\mu_0 i}{2\pi r}$ i.e., $B \propto \frac{1}{r}$
which implies that field has cylindrical symmetry.

48. (a) 49. (d) 50. (c)

51. (d) There is no current inside the pipe. Therefore

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I$$

$$I = 0 \quad \therefore B = 0$$

52. (c) 53. (b) 54. (a)

55. (d) $B = \mu_0 n I = 4\pi \times 10^{-7} \times 10 \times 5 = 2\pi \times 10^{-5} \text{ T}$.

56. (b) Magnetic field at a point on one end of a solenoid

$$B = \frac{1}{2} \mu_0 n i$$

$$57. (a) \quad \frac{B_2}{B_1} = \frac{\mu_0 n_2 i_2}{\mu_0 n_1 i_1} \Rightarrow \frac{B_2}{6.28 \times 10^{-2}} = \frac{100 \times \frac{3}{2}}{200 \times i}$$

$$\Rightarrow B_2 = \frac{6.28 \times 10^{-2}}{6} = 1.05 \times 10^{-2} \text{ Wb/m}^2$$

$$58. (d) \quad B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi n i_1}{r_1} - \frac{\mu_0}{4\pi} \cdot \frac{2\pi n i_2}{r_2} = \frac{\mu_0}{2} \left[\frac{n i_1}{r_1} - \frac{n i_2}{r_2} \right]$$

59. (b) Let ℓ be length of wire.

$$\text{1st case : } \ell = 2\pi r \Rightarrow r = \frac{\ell}{2\pi}$$

$$B = \frac{\mu_0 I n}{2r} = \frac{\mu_0 I \times 2\pi}{2\ell} = \frac{\mu_0 \pi I}{\ell} \quad [\because n = \frac{\ell}{2\pi}] \dots (1)$$

$$\text{2nd Case : } \ell = 2(2\pi r') \Rightarrow r' = \frac{\ell}{4\pi}$$

$$B' = \frac{\mu_0 I n}{2 \cdot \frac{\ell}{4\pi}} = \frac{2\mu_0 I \pi}{\ell} = 4 \left(\frac{\mu_0 \pi I}{\ell} \right) = 4B$$

60. (b) We know that magnetic field at the centre of circular coil,

$$B = \frac{\mu_0 I n}{2r} = \frac{4\pi \times 10^{-7} \times 2 \times 50}{2 \times 0.5} = 1.25 \times 10^{-4} \text{ T}$$

61. (a) A current carrying coil has magnetic dipole moment.

Hence a torque $\vec{p}_m \times \vec{B}$ acts on it in magnetic field.

62. (a)

63. (d) $\tau = MB \sin \theta \Rightarrow \tau_{\max} = NIAB$, ($\theta = 90^\circ$)

64. (c) The magnetic field is perpendicular to the plane of the paper. Let us consider two diametrically opposite elements. By Fleming's left hand rule, on element AB the direction of force will be leftwards and the magnitude will be

$$dF = I(d\ell)B \sin 90^\circ = I(d\ell)B$$

On element CD , the direction of force will be towards right on the plane of the paper and the magnitude will be dF

$$= I(d\ell)B.$$

These two forces will cancel out.

65. (a) Force between two long conductor carrying current,

$$F = \frac{\mu_0}{4\pi} \frac{2I_1 I_2}{d} \times \ell; \quad F' = -\frac{\mu_0}{4\pi} \frac{2(2I_1)I_2}{3d} \ell$$

$$\therefore \frac{F'}{F} = \frac{-2}{3}$$

66. (a) $F = B i l \sin \theta \Rightarrow 7.5 = 2 \times 5 \times 1.5 \sin \theta \Rightarrow \theta = 30^\circ$ 67. (b) $F = B i l = 2 \times 1.2 \times 0.5 = 1.2 \text{ N}$

$$68. (a) \quad F' = \frac{\mu_0 i_1 i_2}{2\pi r} = \frac{4\pi \times 10^{-7} \times 30 \times 30}{2 \times \pi \times 5 \times 10^{-2}} = 3.6 \times 10^{-3} \text{ N/m}.$$

$$69. (a) \quad \frac{F}{\ell} = \frac{\mu_0 i^2}{2\pi d} = 9.8 \times 4 \times 10^{-6}$$

$$\Rightarrow i = \sqrt{\frac{4 \times 10^{-6} \times 9.8 \times 0.12}{2 \times 10^{-7}}} = 4.85 \text{ A}$$

70. (b) $F = iB l \sin \theta$. This is maximum when $\sin \theta = 1$ or $\theta = \pi/2$.71. (b) $F = I \ell B \sin \theta = 3 \times 0.40 \times (500 \times 10^{-4}) \times \sin 30^\circ = 3 \times 10^{-2} \text{ N}$.

72. (d) At these points, the resultant field = 0

$$73. (a) \quad F = \frac{\mu_0}{4\pi} \times \frac{2i_1 i_2}{r}$$

$$= 50 \times 10^{-7} \text{ N/m. Here } F \text{ is force per unit length.}$$

74. (b) Current carrying conductors will attract each other, while electron beams will repel each other.

75. (b) Magnetic field at a point on one end of a solenoid

$$B = \frac{1}{2} \mu_0 n i$$

76. (b)

77. (b) The magnetic field from the centre of wire of radius R is given by

$$B = \left(\frac{\mu_0 I}{2R^2} \right) r \quad (r < R) \Rightarrow B \propto r$$

$$\text{and } B = \frac{\mu_0 I}{2\pi r} \quad (r > R) \Rightarrow B \propto \frac{1}{r}$$

From the above descriptions, we can say that the graph (b) is a correct representation.

Moving Charges and Magnetism

78. (a) $B = \frac{\mu_0 I}{2r} \times \frac{\theta}{2\pi} = \frac{\mu_0 I \theta}{4\pi r}$

79. (c) Since n is an even number, we can assume the wires in pairs such that the two wires forming a pair is placed diametrically opposite to each other on the surface of cylinder. The fields produced on the axis by them are equal and opposite and can get cancelled with each other.

80. (b)

81. (c) As Lorentz force is given by

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}) = q\vec{E} + q(\vec{v} \times \vec{B})$$

$$\vec{F} = \vec{F}_E + \vec{F}_B$$

82. (a) Lorentz force, $\vec{F} = q\{\vec{E} + (\vec{v} \times \vec{B})\}$

$$\vec{v} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 0 \\ 5 & 3 & 4 \end{vmatrix} = 8\hat{i} - 4\hat{j} - 7\hat{k}$$

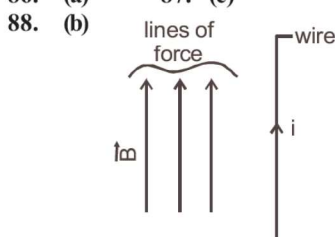
$$\vec{F} = 1(2\hat{i} - 3\hat{j} + 8\hat{i} - 4\hat{j} - 7\hat{k}) = (10\hat{i} - 7\hat{j} - 7\hat{k})$$

83. (c)

84. (c) $E = vB = 2 \times 10^3 \times 1.5 = 3 \times 10^3 \text{ V/m}$.

85. (b) $\tau = m\beta \sin \theta = 0.75 \times 0.20 = 0.15 \text{ Nm}$

86. (a) 87. (c)

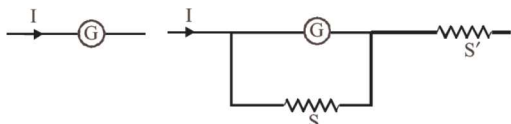


$F_m = I(\vec{l} \times \vec{B}) = 0$ because the directions of current $\vec{l}$ and magnetic field $\vec{B}$ are parallel.

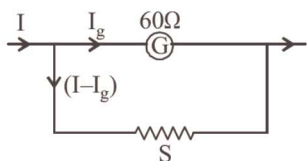
89. (d)

90. (c) To keep the main current in the circuit unchanged, the resistance of the galvanometer should be equal to the net resistance.

$$\therefore G = \left(\frac{GS}{G+S} \right) + S' \Rightarrow G - \frac{GS}{G+S} = S' \quad \therefore S' = \frac{G^2}{G+S}$$



91. (c) $G = 60\Omega$, $I_g = 1.0\text{A}$, $I = 5\text{A}$.



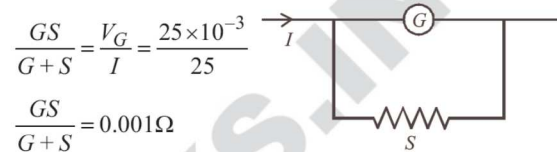
Let S be the shunt resistance connected in parallel to galvanometer

$$I_g G = (I - I_g) S,$$

$$S = \frac{I_g G}{I - I_g} = \frac{1}{5 - 1} \times 60 = 15\Omega$$

Thus by putting 15Ω in parallel, the galvanometer can be converted into an ammeter.

92. (a) Galvanometer is converted into ammeter, by connected a shunt, in parallel with it.



$$\frac{GS}{G+S} = \frac{V_G}{I} = \frac{25 \times 10^{-3}}{25}$$

$$\frac{GS}{G+S} = 0.001\Omega$$

$$\text{Here } S \ll G \text{ so } S = 0.001\Omega$$

93. (a) $i_g = i \frac{SG}{S+G}$

$$\text{or } 100 \times 10^{-6} = i \left(\frac{0.1 \times 100}{100 + 0.1} \right) \text{ or } i = 100.1 \times 10^{-3} \text{ A}.$$

94. (c) $I_g = 10 \times 10^{-3} / 5 = 2 \times 10^{-3} \text{ A}$;

$$R = \frac{V}{I_g} - R_g = \frac{1}{2 \times 10^{-3}} - 5 = 495\Omega.$$

(Sol. 95-99):

$$\text{We know that } r = \frac{mV}{qB}$$

$$\therefore V > \frac{qBl}{m}$$

95. (a) If $r > l$ then particle enter the III region $\frac{mV}{qB} > l$

96. (c) If $V = \frac{qBl}{m}$ then particle will cover semi circular path in this condition the path length of the particle in region II is maximum.

97. (d) Time spent in region II, $T = \frac{\pi m}{qB}$

It does not depends upon the velocity.

98. (d)

99. (d) $F = q(\vec{V} \times \vec{B})$ if $V \parallel B$, then $\vec{F} = 0$

100. (d) Ampere's circuital law can be derived from Biot-Savart law

101. (c) The magnetic field of two equal halves of the loop is equal and opposite and so $\vec{B} = 0$.

102. (a) 103. (b)

104. (a) A moving charge experiences a force in magnetic fields. It is because of interaction of two magnetic fields, one which is produced due to the motion of charge and other in which charge is moving.

105. (c) Due to electric field, the force is $\vec{F} = q\vec{E}$ in the direction of $\vec{E}$. Since $\vec{E}$ is parallel to $\vec{B}$, the particle velocity $\vec{v}$ (acquired due to force $\vec{F}$) is parallel to $\vec{B}$. Hence $\vec{B}$ will not exert any force since $\vec{v} \times \vec{B} = 0$ and the motion of the particle is not affected by $\vec{B}$.

106. (c)

107. (c) Since the magnetic force is always perpendicular to the velocity of the charged particle so, work done is always zero.

108. (d)

109. (d) The magnetic force acts on moving electrons, and so net force on the conductor is non-zero.

110. (b) If $\vec{F} \perp \vec{v}$ at all instants then motion will be circular

111. (c)

112. (a) (A) $\rightarrow$ (4); (B) $\rightarrow$ (3); (C) $\rightarrow$ (1); (D) $\rightarrow$ (2)

113. $(\vec{F}_E + \vec{F}_B)$ ($\vec{F} = \vec{F}_E + \vec{F}_B$ acting on)

As Lorentz force is given by

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}) = q\vec{E} + q(\vec{v} \times \vec{B})$$

$$\vec{F} = \vec{F}_E + \vec{F}_B$$

114. $(11q\hat{j})$ Lorentz force acting on the particle

$$\vec{F} = q[\vec{E} + \vec{v} \times \vec{B}]$$

$$= q \left[3\hat{i} + \hat{j} + 2\hat{k} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 4 & 1 \\ 1 & 1 & -3 \end{vmatrix} \right]$$

$$= q[3\hat{i} + \hat{j} + 2\hat{k} + \hat{i}(-12-1) - \hat{j}(-9-1) + \hat{k}(3-4)]$$

$$F_y = 11q\hat{j}$$

115. $(\vec{m} \times \vec{B})$

116. (0.24) $F = IIB$

Here, $\theta = 90^\circ$, $I = 10 \text{ A}$

$$I = 8 \text{ cm} = 8 \times 10^{-2} \text{ m}, B = 0.3 \text{ T}$$

$$\therefore F = 10 \times 8 \times 10^{-2} \times 0.3 \times \sin 90^\circ = 0.24 \text{ N}$$

117. (3.33×10^{-4}) $B = \frac{\mu_0 I}{2\pi r} \Rightarrow B \propto \frac{1}{r}$

As the distance is increased to three times, the magnetic

induction reduces to one third. Hence, $B = \frac{1}{3} \times 10^{-3} \text{ tesla}$

$$= 3.33 \times 10^{-4} \text{ tesla}$$

118. (False) The magnitude of magnetic force on charged particle undergoing uniform circular motion in uniform magnetic field is $F = qvB$

$\therefore$ If v is doubled keeping q and B constant, the force F just doubles. Hence statement-1 is false.

119. (False)

120. (True) The magnitude of magnetic field is different at different points.

121. (True) As magnetic field and velocity are in same direction.

122. (True)

123. (True)